(PV-CSRS-01) SwRS for Cybersecurity

Document ID: `PV-CSRS-01` Product: `Portview` Document Status: `Released`

Document Approval

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Revision History

Rev.	Date	Description
`0.0`	`2023-12-11`	Initial version based on requirements subdivision
`0.1`	`2025-07-24`	Revised in compliance with MDCG 2019-16 Rev.1 Guidance on Cybersecurity for medical devices

1. Purpose

This document covers the cybersecurity risk assessment report of the intraoral imaging system named Portview in the software lifecycle process.

It contains:

- the risk analysis
- the risk assessment report
- the risk traceability matrix with testable requirements

2. Referenced Documents

#	Standard No.	Standard Title
1	EN ISO 13485:2016	Medical devices - Quality management system - Requirements for regulatory purposes
2	EN ISO 14971:2019	Medical devices - Application of risk management to medical devices
3	EN 62304:2006/A1:2015	Medical device software - Software life-cycle processes
4	AAMI TIR 57:2016/(R)2019	Principles for medical device security - Risk management
5	Medical Device Regulation (EU) 2017/745	Medical Device Regulation
6	MDCG 2019-16 Rev.1	Guidance on Cybersecurity for medical devices

3. Device Description

3.1 Device Information

- **Product Name**: Intraoral imaging system
- **Software Name**: Portview
- **Manufacturer**: GENORAY Co., Ltd.

3.2 Intended Use

It is intended for use in dental applications for the scanning and processing of image data in an intraoral imaging system.

3.3 User Environment

It is to be used in health care facilities both inside and outside the operating room in a variety of procedures.

- Diagnosis and monitoring of dental health disease conditions

4. Basic Information

4.1 Health Data (PII)

- Patient digital image for diagnosis
- Patient ID (assigned by the program, not a social security number)
- Patient Name
- Patient Sex (optional)
- Patient Birth Date and Age (optional)

4.2 Functions

- **Transmission protocols**: TCP-based Private Network
- **Transport protocols**: TCP, NBT, NBF
- **Transmission data**: Structured Query Language Data, Medical Imaging Information and Related Data

4.3 Uses

Clinical application: Digital intraoral image acquisition workstation

4.4 Interchange Ways (System Requirements)

Item	Specification
OS	Microsoft Windows 10 or higher (32-bit or 64-bit)
CPU	Core Duo / Core2 Processor
Memory	RAM 4 GB or more
VGA	Onboard chipset or more
Storage	100 GB free hard disk space or more
Network	100/1000 Mbps Ethernet
ODD	DVD-Multi
Graphic requirement	32-bit color display

Item	Specification
Monitor resolution	1280 x 1024 or more

4.5 Security Software

None

4.6 Off-The-Shelf Software

Type	Model	Full Version	Supplier
Operating system	Microsoft Windows 10 IoT	1809	Microsoft Corporation
Toolkit	Windows SDK	10.0.17763.0	Microsoft Corporation
Toolkit	DCMTK	3.6.7	OFFIS
Development framework	Qt	5.13	The Qt Company

5. Risk Management

5.1 Cybersecurity Essentials Checklist

Checklist Requirements

No.	Classification	Risk Item	Requirement	Applicability
1	Identify / Protect	Limit access to trusted users only	The software has account management system	Yes
2	Identify / Protect	Limit access to trusted users only	The software provides a means to block unauthorized network communication	Yes

No.	Classification	Risk Item	Requirement	Applicability
3	Identify / Protect	Ensure trusted content	The system shall provide a means for users to obtain administrator approval to make software or firmware updates	Yes
4	Identify / Protect	Ensure trusted content	Use systematic procedures for authorized users to download version-identifiable software and firmware from the manufacturer	N/A
5	Identify / Protect	Ensure trusted content	The system shall ensure confidentiality and integrity by using appropriate encryption and decryption methods when exchanging medical device control information through the network	N/A
6	Identify / Protect	Ensure trusted content	It is recommended not to store personal medical information in measuring devices or gateways used outside of medical institutions	N/A

No.	Classification	Risk Item	Requirement	Applicability
7	Detect / Respond / Recover	System logging for data auditing	The system shall provide a means to notify the user when a business process is interrupted or an error occurs	Yes
8	Detect / Respond / Recover	System logging for data auditing	The software shall provide a means to support the audit trail	Yes
9	Detect / Respond / Recover	Provide information	Manufacturers shall define countermeasures to be taken when detecting cybersecurity threats and provide relevant information to end users	N/A
10	Detect / Respond / Recover	Protect critical functionality	The systems shall implement functions that protect core information in the event of a cybersecurity breach	N/A

Checklist Risk And Traceability

No.	Risk	RS (`PV-SRS-01`)	SDS (`PV-SDS-01`)	V&V (`PV-TP-01`)
1	PV-01, PV-02, PV-03, PV-04	`SRS-001` to `SRS-005`	`SDS-001`	`ITP-002` to `ITP-005`
2	PV-50	`SRS-034`	`SDS-020`	`ITP-034`
3	PV-51	`SRS-001`	`SDS-001`	`ITP-036`

No.	Risk	RS (`PV-SRS-01`)	SDS (`PV-SDS-01`)	V&V (`PV-TP-01`)
4	Remote installer download is not supported	—	—	—
5	P2P-only; secure transmission guaranteed	—	—	—
6	PV-50; private networks only	—	—	—
7	PV-01, PV-02, PV-03, PV-04	`SRS-001` to `SRS-005`	`SDS-001`	`ITP-002` to `ITP-005`
8	PV-52	`SRS-035`	`SDS-021`	`ITP-035`
9	P2P-only; secure transmission guaranteed	—	—	—
10	P2P-only; secure transmission guaranteed	—	—	—

5.2 Identification Of Cybersecurity Risk

Refer to Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

5.3 Cybersecurity Risk Evaluation

Refer to Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

5.4 Cybersecurity Risk Control

Refer to Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

5.5 Risk Control Verification

See section 5.1 Cybersecurity Essentials Checklist in this document and Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

5.6 Risk/Benefit Analysis

Refer to Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

5.7 Overall Residual Risk Evaluation

Overall residual risk is re-evaluated after the cybersecurity risk controls have been applied.

Refer to Clause 12, Risk Analysis Table (FMEA) in the Risk Management Report. See also `FMEA-Z01` Portview Risks FMEA.

6. Maintenance Plan

Computer systems require regular maintenance in order to continue running smoothly and with proper security. Each vendor (or user) will be informed about maintenance and security via email.

A help desk team will start supporting users right away to take care of their immediate needs. The team will contact vendors (or users) for security patches if a critical threat to system operation is found.

The maintenance planning activities are performed in the software validation maintenance process in accordance with IEC 62304:2015. Refer to `PV-SV-02` Section 11 for the full maintenance and problem-resolution process.

6.1 Security Patch Priority And Deployment

Priority	Criteria	Deployment Timeline
Critical	Confirmed exploitable vulnerability affecting patient data, system availability, or regulatory compliance	Immediate; expedited change-control and verification
High	Confirmed vulnerability with limited exploit potential or indirect patient-safety impact	Within 30 days through expedited change-control
Normal	Advisory or preventive update without confirmed exploit	Next scheduled release cycle

Security patches follow the change-control and verification process defined in `PV-SV-02` Section 11.2. SOUP components (Windows SDK, DCMTK, Qt) are monitored for end-of-life advisories and patched according to the same priority scheme.